

Md Jueal Mia

PH.D. CANDIDATE · COMPUTER SCIENCE

Knight Foundation School of Computing and Information Sciences (KFSCIS), Florida International University, Miami, FL-33199

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Research Interests

Trustworthy AI, AI Security, Foundation Models, Large Language Models, Agentic AI, Federated Learning, Privacy-Preserving Machine Learning

Selected Highlights

- Published **7 peer-reviewed publications** (6 first-author; 3 journal articles and 4 conference papers), with **5 additional manuscripts under review** (4 journal articles and 1 conference papers).
- Submitted **3 U.S. patent applications** in trustworthy AI, federated learning, and foundation models.
- Contributed to multiple successful externally funded research proposals and projects, including **USDOT** and **NAIRR** initiatives in trustworthy AI, cybersecurity, and intelligent transportation systems.
- Recipient of the **Best Paper Award** at the *Third International Conference on Smart Systems: Innovations in Computing (SSIC 2021)*.

Education

Florida International University

Miami, FL, USA

PH.D. IN COMPUTER SCIENCE

Jan 2023 – Present

- CGPA: 4.00/4.00

Jahangirnagar University

Dhaka, Bangladesh

M.SC. IN COMPUTER SCIENCE AND ENGINEERING

2014 – 2015

Jahangirnagar University

Dhaka, Bangladesh

B.SC. IN COMPUTER SCIENCE AND ENGINEERING

2010 – 2014

Professional Experience

Department of Computer Science, Florida International University

Miami, FL, USA

GRADUATE RESEARCH ASSISTANT

May 2024 – Present

Department of Computer Science, Florida International University

Miami, FL, USA

GRADUATE TEACHING ASSISTANT

Jan 2023 – Apr 2024

Department of Computer Science, University of Nevada, Reno

Reno, NV, USA

GRADUATE RESEARCH ASSISTANT

Aug 2022 – Dec 2022

Department of CSE, Daffodil International University

Dhaka, Bangladesh

ASSISTANT PROFESSOR (STUDY LEAVE)

Feb 2022 – Jul 2022

Department of CSE, Daffodil International University

Dhaka, Bangladesh

LECTURER (SENIOR SCALE)

Feb 2020 – Jan 2022

Teaching Experience

Break Through Tech AI, Cornell Tech

New York, NY, USA

COURSE SUPPORT – MACHINE LEARNING FOUNDATIONS PROGRAM

May 2026 – Present

- Support instruction for the Machine Learning Foundations Program, helping students develop strong foundations in machine learning and artificial intelligence.
- Mentor students through project-based learning, coding exercises, office hours, and technical discussions.
- Evaluate programming assignments, quizzes, and projects while providing constructive feedback to improve student learning outcomes.
- Provide technical guidance on machine learning algorithms, model evaluation, and responsible AI practices in a collaborative learning environment.

Florida International University

Miami, FL, USA

GRADUATE TEACHING ASSISTANT

Jan 2023 – Apr 2024

- Supported instruction for undergraduate courses in Systems Programming and Java Programming.
- Mentored more than 10 undergraduate students through coursework, programming assignments, and technical projects.
- Mentored two NSF REU research interns on projects in trustworthy AI, federated learning, and secure machine learning.
- Facilitated hands-on learning activities focused on AI security, privacy-preserving machine learning, and large language models.

Daffodil International University

Dhaka, Bangladesh

ASSISTANT PROFESSOR

Feb 2022 – Jul 2022

- Delivered undergraduate instruction, advised students academically, and supervised technical research projects.
- Supervised more than 50 undergraduate capstone and research projects spanning artificial intelligence, machine learning, and mobile application development.

Daffodil International University

Dhaka, Bangladesh

LECTURER (SENIOR SCALE)

May 2016 – Jan 2022

- Taught undergraduate courses to more than 500 students in Analysis of Algorithms, Software Engineering, Java, Python, and Machine Learning.
- Designed and developed the undergraduate curriculum for Analysis of Algorithms.
- Designed and developed the undergraduate curriculum for Mobile Application Development.
- Provided academic advising and mentored undergraduate students on technical projects and research.

Publications

- **7 peer-reviewed publications** (6 first-author; 3 journal articles and 4 conference papers), with **5 manuscripts under review** (4 journal articles and 1 conference papers), under review **3 U.S. patent applications**.

JOURNAL PUBLICATIONS

1. **Md Jueal Mia**; M. Hadi Amini. (2025). QuanCrypt-FL: Quantized Homomorphic Encryption with Pruning for Secure Federated Learning. *IEEE Transactions on Artificial Intelligence*. doi: 10.1109/TAI.2025.3612906.
2. **Md Jueal Mia**; M. Hadi Amini. (2025). BART-FL: A Backdoor Attack-Resilient Federated Aggregation Technique for Cross-Silo Applications. *IEEE Transactions on Machine Learning in Communications and Networking*, vol. 3, pp. 1311–1325. doi: 10.1109/TMLCN.2025.3611398.
3. **Md Jueal Mia**; M. Hadi Amini. (2024). A Secure Object Detection Technique for Intelligent Transportation Systems. *IEEE Open Journal of Intelligent Transportation Systems*, vol. 5, pp. 495–508.

CONFERENCE PUBLICATIONS

1. **Md Jueal Mia**; M. Hadi Amini. (2025). An Empirical Analysis of Secure Federated Learning for Autonomous Vehicle Applications. *In Proceedings of Computing in Civil Engineering 2024*, pp. 417–426.

2. **Md Jueal Mia**; M. Hadi Amini. (2025). JaiLIP: Jailbreaking Vision-Language Models via Loss Guided Image Perturbation. *In Proceedings of the 2025 IEEE International Conference on Machine Learning and Applications (ICMLA)*, Boca Raton, FL, USA, pp. 1362–1368. doi: 10.1109/ICMLA66185.2025.00208.
3. Badhan Chandra Das; Md Tasnim Jawad; **Md Jueal Mia**; M. Hadi Amini; Yanzhao Wu. (2025). Jailbreaking Large Vision-Language Models in Intelligent Transportation Systems. *In Proceedings of the 2025 IEEE International Conference on Machine Learning and Applications (ICMLA)*, Boca Raton, FL, USA, pp. 1423–1429. doi: 10.1109/ICMLA66185.2025.00217.

ACCEPTED CONFERENCE PAPERS

1. **Md Jueal Mia**, M. Hadi Amini. Don't Overthink, Don't Underthink: Toward Adaptive Reasoning in Agentic AI. Accepted at *ACM AI Summit 2026*.

JOURNAL ARTICLES UNDER REVIEW

1. **Md Jueal Mia**; M. Hadi Amini. (2025). FedShield-LLM: A Secure and Scalable Federated Fine-Tuned Large Language Model. Submitted to *IEEE Transactions on Dependable and Secure Computing (TDSC)*.
2. M. Hadi Amini; **Md Jueal Mia**; Yasaman Saadati; Ahmed Imteaj; Seyedsina Nabavirazavi; Urmish Thakker; Md Zarif Hossain; Awal Ahmed Fime; S. S. Iyengar. (2025). Distributed LLMs and Multimodal Large Language Models: A Survey on Advances, Challenges, and Future Directions. *arXiv:2503.16585*. arXiv.
3. Hojat Allah Salehi; **Md Jueal Mia**; S. Sandeep Pradhan; M. Hadi Amini; Farhad Shirani. (2024). CorBin-FL: A Differentially Private Federated Learning Mechanism using Common Randomness. Submitted to *IEEE Open Journal of the Communications Society*.
4. Ahmed Aljohani; **Md Jueal Mia**; M. Hadi Amini. Backdoor-Resilient Lane Detection for Autonomous Vehicles: Assessment and Practical Defenses. Submitted to *IEEE Transactions on Intelligent Transportation Systems*.

CONFERENCE PAPERS UNDER REVIEW

1. **Md Jueal Mia**, Joaquin Molto, Yanzhao Wu, M. Hadi Amini. GUARD-SLM: Token Activation-Based Defense Against Jailbreak Attacks for Small Language Models. Submitted to *ACL ARR 2026*.

PATENTS

1. Mohammadhadi Amini, **Md Jueal Mia**. *A Privacy-Preserving Federated Fine-Tuned Large Language Model*. Submitted to the U.S. Patent and Trademark Office (USPTO), Pending, Serial No. 19/272,726, 2025.
2. Mohammadhadi Amini, **Md Jueal Mia**. *Systems and Methods for Adversarial and Jailbreak Attack-Resilient Federated Fine-Tuned Multimodal Large Language Models*. Patent Application Under Review.
3. Mohammadhadi Amini, **Md Jueal Mia**, Yanzhao Wu. *Systems and Methods for Defense against Jailbreak Attacks in Small Language Models*. Patent Application Under Review.

Presentations

CONTRIBUTED PRESENTATIONS

- **Md Jueal Mia** and M. Hadi Amini. "A Privacy-aware Cyber-attack-resilient Image Recognition Algorithm for Autonomous Vehicles." Poster presentation at the National Center for Transportation Cybersecurity and Resiliency (TraCR) Annual Conference, Greenville, SC, USA, May 6, 2024.

Research Funding & Proposal Development

COLLABORATIVE PROPOSAL CONTRIBUTIONS

- Contributed to the development of the successful USDOT National Center for Transportation Cybersecurity and Resiliency (TraCR) proposal, *"Secure and Privacy-Preserving Federated Learning for Connected and Automated Vehicles."*
- Contributed to the proposal development and technical research for *"Empirical Security Analysis and Lightweight Defense Mechanisms Against Jailbreak Attacks in Small Language Models on Edge Devices"*, funded through the National Artificial Intelligence Research Resource (NAIRR) Pilot (Grant No. NAIRR260008), supporting **3,000 Purdue Anvil GPU hours**.
- Contributed to the proposal development and technical research for *"Empirical Security Analysis and Defense Mechanisms Against Jailbreak Attacks in Large Language Models"*, funded through the National Artificial Intelligence Research Resource (NAIRR) Pilot (Grant No. NAIRR260027), supporting **\$4,500 in CloudBank credits**.

RESEARCH GRANTS AND STUDENT MENTORING

- Co-authored an NSF REU proposal, *"Enhancing Security Issues in Federated Learning for Resource-Constrained IoT Device Applications,"* and co-supervised undergraduate researcher Hannan Shah.
- Co-authored an NSF REU proposal, *"Enhancing Security Issues in Machine Learning for Application Development,"* and co-supervised undergraduate researcher Isabella Peng.

Research News Coverage

- **FIU News.** *"FIU Researchers Reveal How Altered Images Can Bypass AI Safeguards."* July 2026.
- **USA Today.** *"You can use altered images to trick AI systems, Florida study shows"* July 2026.
- **Yahoo Finance.** *"Florida International University Researchers Reveal How Altered Images Can Bypass AI Safeguards."* July 2026.
- **Sun Sentinel.** *"FIU Researchers Find New Weakness in AI Chatbots as Lawsuits Grow."* July 2026.

Selected Research and Project Collaborators

Florida International University (FIU): M. Hadi Amini (Associate Professor), Kemal Akkaya (Professor), Farhad Shirani (Assistant Professor), S. S. Iyengar (Distinguished University Professor), Yanzhao Wu (Assistant Professor), and Selcuk Uluagac (Eminent Scholar Chaired Professor).

Purdue University: Satish Ukkusuri (Professor) and Yiheng Feng (Assistant Professor).

University of Alabama: Mizanur Rahman (Associate Professor).

Southern Illinois University Carbondale: Ahmed Imteaj (Assistant Professor).

Morgan State University: Mansoureh Jeihani (Professor).

SambaNova Systems / University of Wisconsin: Urmish Thakker (Deep Learning Researcher).

Professional Service

EDITORIAL AND REVIEW ACTIVITIES

Journal Reviewer

- *IEEE Transactions on Information Forensics and Security (TIFS)* — **Impact Factor: 6.3**
- *IEEE Transactions on Dependable and Secure Computing (TDSC)* — **Impact Factor: 6.8**
- *IEEE Transactions on Vehicular Technology (TVT)* — **Impact Factor: 6.1**

- *IEEE Transactions on Consumer Electronics (TCE)* — **Impact Factor: 4.3**
- *IEEE Transactions on Machine Learning in Communications and Networking (TMLCN)*
- *Springer Nature Journals*

Conference Reviewer

- IEEE International Conference on Distributed Computing Systems (ICDCS 2024)
- International Conference on Machine Intelligence and Data Science Applications (MIDAS 2021)
- International Joint Conference on Advances in Computational Intelligence (IJCACI 2020)

SERVICE

- Program Committee Member, *IEEE International Conference on Tools with Artificial Intelligence (ICTAI)*, 2026.
- Organizing Committee Member, *International Joint Conference on Advances in Computational Intelligence (IJCACI)*, 2020.
- Organizing Committee Member, *International Collegiate Programming Contest (ICPC)*, 2018.

PROFESSIONAL DEVELOPMENT

- Faculty Research Training, *Road to Research*, Daffodil International University, 2018.
- Faculty Workshop on *Machine Learning and AI*, Daffodil International University, 2018.
- Faculty Workshop on *Research Proposal Development*, Daffodil International University, 2017.
- Participatory Engaging Techniques (PET) Teacher Training, Daffodil International University, 2017.

Awards, Fellowships, & Grants

- **Publication Grant** Recipient for Journal from the University Graduate School, Florida International University (2024, 2025).
- **Travel Grant** Recipient for Conference from the University Graduate School and Knight Foundation School of Computing and Information Sciences (KFSCIS), Florida International University (2024, 2025).
- **Best Paper Award** at the Third International Conference on Smart Systems: Innovations in Computing (SSIC), Springer (2021).
- In Recognition of Scholarly Publication in Reputed Indexed Journals for the year by Daffodil International University. (2020, 2021, 2022)
- Received **Best Performer-2018** by the Department of Computer Science and Engineering, Daffodil International University, for outstanding academic and professional contributions (2018).
- University Scholarship for Outstanding Academic Performance, awarded annually for top academic performance (2010–2014).
- Undergraduate Entrance Exam Scholarship for achieving the top position in the university entrance exam (2010–2011).